METAL SOLUTIONS

EOS StainlessSteel 316L-4404

Material Data Sheet

EOS STAINLESSSTEEL 316L-4404

EOS StainlessSteel is a high-performance, marine-grade austenitic stainless steel that is molybdenum alloyed for enhanced corrosion resistance in chloride environments. 316L is a standard material for numerous applications in process, energy, paper, transportation, and other industries. EOS Stainless Steel 316L-4404 has a chemical composition corresponding to DIN EN 10088-3. All EOS StainlessSteel 316L parameter sets, including 40/80 µm, are compatible with EOS StainlessSteel 316L-4404 powder.

MAIN CHARACTERISTICS

- High ductility and toughness
- High strength
- High corrosion resistance

TYPICAL APPLICATIONS

- Chemical industry
- Food processing
- Water handling and marine applications

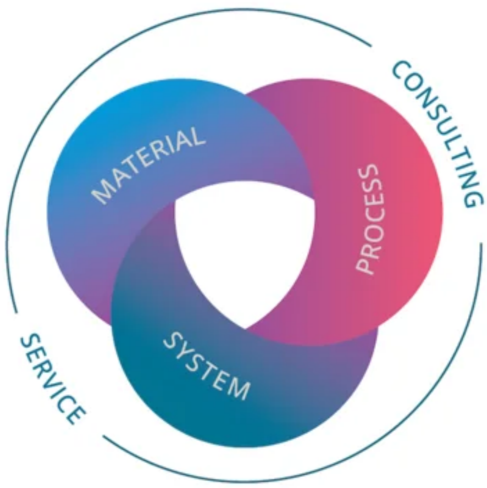
The EOS Quality Triangle

EOS uses an approach that is unique in the AM industry, taking each of the three central technical elements of the production process into account: the system, the material and the process. The data resulting from each combination is assigned a Technology Readiness Level (TRL) which makes the expected performance and production capability of the solution transparent.

EOS incorporates these TRLs into the following two categories:

- Premium products (TRL 7-9): offer highly validated data, proven capability and reproducible part properties.
- Core products (TRL 3 and 5): enable early customer access to newest technology still under development and are therefore less mature with less data.

All of the data stated in this material data sheet is produced according to EOS Quality Management System and international standards

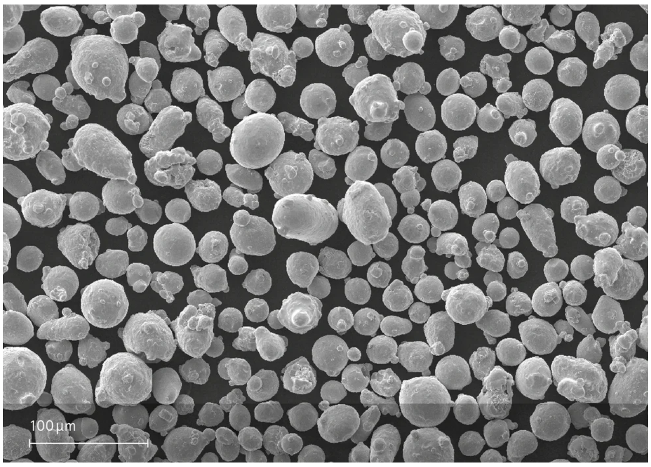


POWDER PROPERTIES

The chemical composition of EOS stainlessSteel 316L-4404 is in compliance with DIN EN 10088-3.

Powder Chemical Composition (wt.-%)

Element	Min.	Max.
Fe	Balance	
Cr	16.5	18.5
Ni	10	13
Mo	2	2.5
C	-	0.03
N	-	0.1
Si	-	1
Mn	-	2



Powder Particle Size

Generic Particle Size Distribution	20 - 65 µm

HEAT TREATMENT

Description

Optional according to DIN EN 10088-3

Steps

Stress relieve: Hold temperature 900 C, hold time minimum 2 h when throughly heated, water quenching or high speed gas quenching. Cooling rate to be high enough to prevent chromium carbide precipitation.

Solution annealing: Hold temperature 1100 C, hold time minimum 1,5 h when throughly heated, water quenching or high speed gas quenching. Cooling rate to be high enough to prevent chromium carbide precipitation.

PROCESS DATA SHEET

EOS StainlessSteel 316L-4404 for EOS M 290 | 40µm

EOS M 290 - 40 µm - TRL 3

Process Information Metal

EOS StainlessSteel 316L-4404 has chemical composition corresponding to DIN EN 10088-3. For more information regarding the powder product, please refer to the EOS StainlessSteel 316L-4404 Material Data Sheet.

The following data was generated using the system and process described below. All EOS StainlessSteel 316L parameter sets, including 40/80 µm, are compatible with EOS StainlessSteel 316L-4404 powder.

System Setup	EOS M 290
EOS Material set	316L_040_FlexM291
Software Requirements	EOSPRINT 2.7 or newer EOSYSTEM 2.11 or newer
Recoater Blade	HSS (High Speed Steel)
Nozzle	EOS Grid Nozzle
Inert gas	Argon
Sieve	63 µm

Additional Information	
Layer Thickness	40 µm
Volume Rate	3.7 mm^3/s
Typical Dimensional Change after HT [%]	+ 0.02 %

Chemical and Physical Properties of Parts

Microstructure of the Produced Parts

Defects	Thickness	Result	Number of Samples
Average Defect Percentage	40 µm	<0.05 %	-

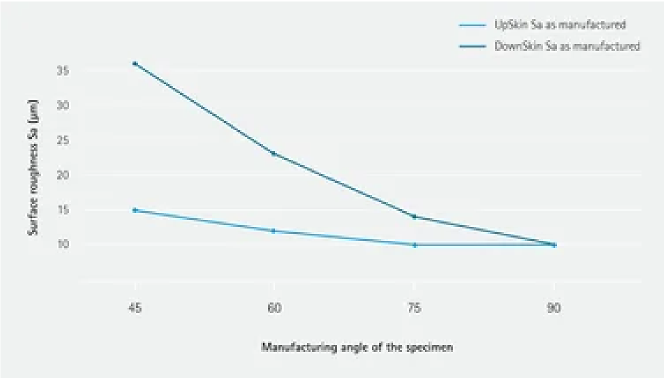
Density EN ISO 3369	Thickness	Result	Number of Samples
Average Density	40 µm	≥7.97 g/cm^3	-

Mechanical Properties

Mechanical Properties As Manufactured

EN ISO 6892-1 | 40 µm	Yield Strength [MPa]	Tensile Strength [MPa]	Elongation at Break A [%]	Reduction of Area Z [%]	Modulus of elasticity [GPa]	Number of Samples
Vertical	490	600	55	-	-	-
Horizontal	560	690	40	-	-	-

Surface Roughness



HEADQUARTERS

EOS GmbH
Electro Optical Systems

Robert-Stirling-Ring 1
82152 Krailling / Munich
Germany

Tel.: +49 89 893 36-0
Email: info@eos.info
URL: www.eos.info

This powder has not been developed, tested or certified as a medical device according to Directive 93/42/EEC (MDD) or Regulation (EU) 2017/745 (MDR) and is not intended to be used as a medical device, in particular for the purposes specified in Art. 2 No. 1 MDR. Insofar as you intend to use the powder as raw material for the manufacture of pharmaceutical products or medical devices (e.g. as raw material which as a material must meet the requirements of Annex 1, Chapter II MDR), the responsibility and liability for all analyses, tests, evaluations, procedures, risk assessments, conformity assessments, approval and certification procedures as well as for all other official and regulatory measures required for this purpose shall lie solely with you both with regard to the pharmaceutical product and/or medical device manufactured by you and with regard to the properties, suitability, testing, evaluation, risk assessment, other requirements for use of the powder as raw material. In this respect, the limitations of liability pursuant to our General Terms and Conditions and the system sales or material contracts shall apply.

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